

Deployment, Operations & Observability

ServiceGuard AI

1. Development deployment

```
Docker Compose
███ frontend
███ backend
███ ai-service
███ postgres
███ redis
```

2. Production evolution

```
Load balancer
↓
Spring Boot replicas
↓
Managed PostgreSQL + Redis
↓
Kafka
↓
AI workers
↓
LLM / model services
```

3. Configuration

- DB_URL/credentials.
- JWT secret.
- Python service URL/token.
- LLM key/model.
- Embedding configuration.
- Redis URL.
- Kafka brokers.
- External integration keys.

4. Observability

- Structured application logs.
- Correlation ID from frontend through Java and Python.
- API latency/error metrics.
- DB/query metrics.
- AI tool-call duration.
- RAG retrieval metrics.
- ML inference metrics.
- Audit event monitoring.

5. Reliability

- Timeouts on downstream calls.
- Bounded retries for transient failures.
- Circuit-breaking where appropriate.

- Idempotent ingestion/incident creation.
- Graceful degradation when AI is unavailable.

6. Backup/recovery

- Regular PostgreSQL backups.
- Migration versioning.
- Model artifact versioning.
- Runbook document/version retention.
- Documented restore procedure.